

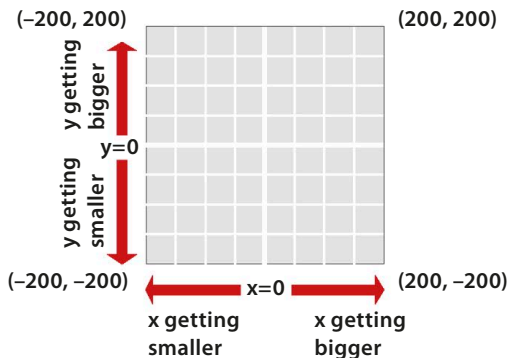
Filling in the blanks

It's time to replace **pass** in the placeholder functions with actual code. After adding the code for each function, run the game to see what difference it makes.

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Stay inside

Fill the `outside_window()` function with this code. First it calculates the position of each of the walls. Then it asks the caterpillar for its current position. By comparing the caterpillar's coordinates with the coordinates of the walls, it can tell whether the caterpillar has left the window. Run the program to check the function works – the caterpillar should stop when it reaches the edge.


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GAME OVER!

Now you know when the caterpillar has left the screen, display a message to tell the player the game has ended. Fill in the `game_over()` function with this code. When called, the function will hide the caterpillar and leaf, and write "GAME OVER!" on the screen.

```
def outside_window():
    left_wall = -t.window_width() / 2
    right_wall = t.window_width() / 2
    top_wall = t.window_height() / 2
    bottom_wall = -t.window_height() / 2
    (x, y) = caterpillar.pos()
    outside = \
        x < left_wall or \
        x > right_wall or \
        y < bottom_wall or \
        y > top_wall
    return outside
```

This function returns two values (a "tuple").

If any of the four conditions above is True, then `outside` is True.

◀ How it works

The centre of the window has the coordinates (0, 0). Since the window is 400 wide, the right wall is half the width from the centre, which is 200. The code gets the left wall's position by subtracting half the width from 0. In other words, 0 - 200, which is -200. It finds the position of the top and bottom walls by a similar method.



```
def game_over():
    caterpillar.color('yellow')
    leaf.color('yellow')
    t.penup()
    t.hideturtle()
    t.write('GAME OVER!', align='center', font=('Arial', 30, 'normal'))
```

The text should be centred (Python uses the US spelling "center").